

Kaggle #06



Linear Regression - A recap

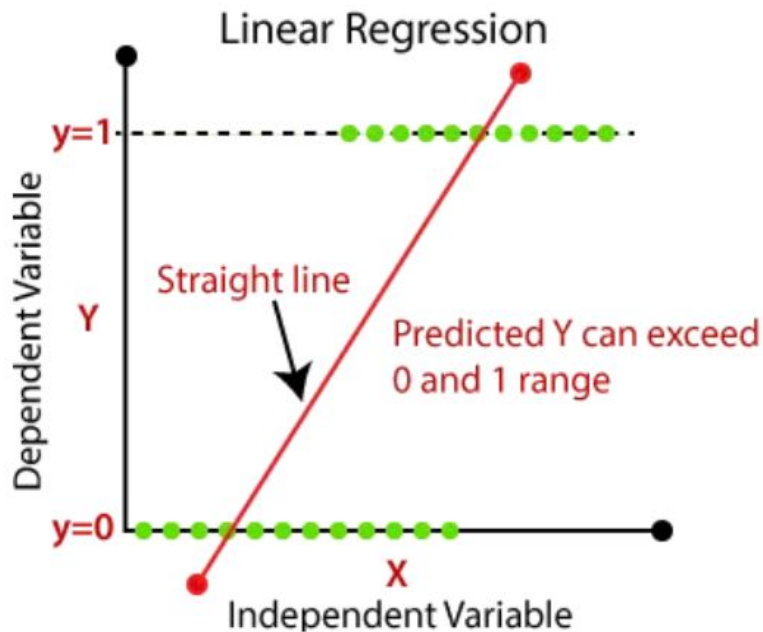
- A **supervised learning** algorithm for **regression tasks**
- Models the relationship between features and a continuous target
- Fits a straight line that best describes the trend in the data
- Works best when changes in inputs cause proportional changes in output
- Commonly used for forecasting and as a baseline model

Examples:

- Predicting house prices, sales figures, or exam scores

“The line of best fit – simple, intuitive, and great for estimating continuous quantities.”

An example of Linear Regression



Linear Regression fits a straight line that best represents how the target changes with input features

How does Linear regression work?

- Prediction is a **weighted sum** of input features
- Learns coefficients that minimise Mean Squared Error (MSE)
- Output is a **continuous numeric value** ($-\infty$ to $+\infty$)
- Assumes a **linear relationship** between inputs and output
- Fits the line so predicted $\hat{y}$ is as close as possible to the true y

Example outputs:

- “How much will this house cost?” → £350 000
- “What will tomorrow’s temperature be?” → 16.7 °C

Adds up weighted feature contributions to predict a single numeric result — any real number.

Shortcomings of Linear Regression

- Outputs are unbounded $\rightarrow$ can be < 0 or $> 1 \rightarrow$ not valid probabilities
- Assumes a linear relationship, which rarely holds for categorical outcomes
- Minimises MSE, not optimised for classification or probability tasks
- Sensitive to outliers and skewed data
- Lacks a way to express uncertainty or class boundaries
- Works for continuous prediction, but not Yes/No decisions

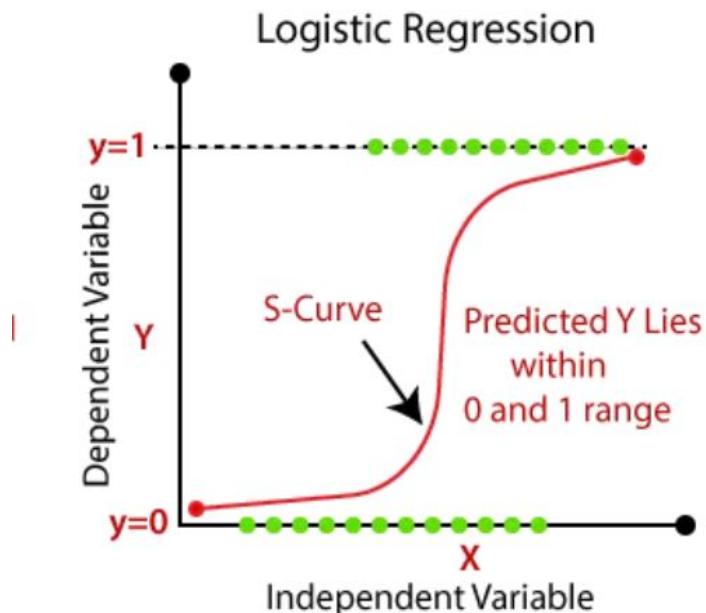
What is Logistic Regression

- A supervised learning algorithm used for binary classification tasks
- Builds on Linear Regression, but predicts probabilities, not raw numbers
- Uses the logistic (sigmoid) function to map predictions into the range $[0, 1]$
- Ideal for Yes/No, True/False, or 0/1 outcomes

Examples:

- Predicting whether an email is spam or not
- Predicting if a loan will be repaid

An example of Logistic Regression



Logistic Regression models the probability of class 1 — mapping any input to a value between 0 and 1

How Logistic Regression Builds on Linear Regression

- Starts with the same linear foundation (weighted sum of inputs)
- Adds the sigmoid (logistic) function to convert outputs into probabilities
- Predicts a probability instead of a raw numeric value
- Uses a different loss function – *Log Loss* instead of *Mean Squared Error*
- Coefficients now describe how each feature affects the log-odds of the outcome

1. Logistic Regression = Linear Regression + Sigmoid + Probability Interpretation

Inputs → Weighted Sum (Linear Model) → Sigmoid Function → Probability Output

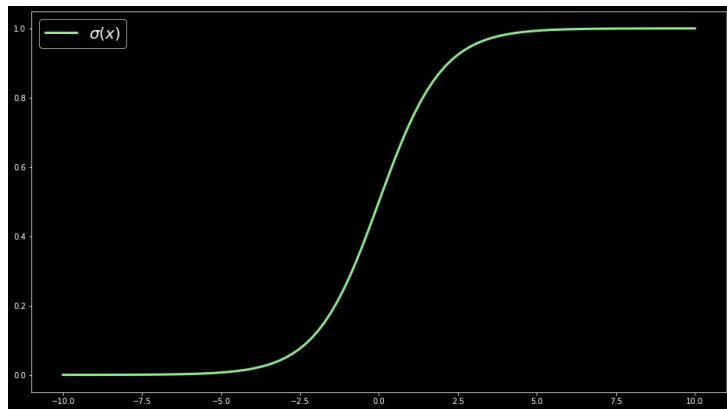
Linear vs Logistic

Concept	Linear Regression	Logistic Regression
Goal	Predict a continuous numeric value	Predict the probability of a class
Output range	$(-\infty, +\infty)$	$[0, 1]$
Task type	Regression	Classification
Output meaning	Predicted numeric value	Probability of being in class
Transformation	None	Sigmoid (logistic)
Loss function	Mean Squared Error (MSE)	Log Loss (Cross-Entropy)
Assumption	Linear relationship between X and Y	Linear relationship between X and log-odds
Interpretation	Each feature changes the predicted value	Each feature changes the odds of the event

Sigmoid

$$\sigma(z) = \frac{1}{1 + e^{-z}}$$

- The **sigmoid function** (also called the **logistic function**) maps any real number to a value between **0** and **1**.
- It's what makes **Logistic Regression** output probabilities instead of raw values.



The sigmoid ‘squashes’ the linear output ($-\infty$ to $+\infty$) into a probability range $[0, 1]$.

Two Ways to View Logistic Regression

- Logistic Regression can be understood in two complementary ways one from a Statistical viewpoint and one from a Machine Learning viewpoint.
- Is it a regression, or classification? -
- BOTH
- It is a regression model, or a classification algorithm
- **Statistical / Theoretical view:** focuses on modelling **probabilities and log-odds**
- **Machine Learning / Practical view:** focuses on **predictions and decision boundaries**

Regression in theory — Classification in practice

Why It's a Regression Model (Statistical View)

- It models the **relationship between features and the probability of an outcome whilst** using a continuous mathematical function.
- It **regresses the log-odds** (logit) of the outcome against the input features.
- The relationship is **linear** in terms of the log-odds, not the probability itself.
- The output is **continuous**, a probability between 0 and 1.
- It belongs to the **Generalised Linear Model (GLM)** family, like Linear Regression.
- No thresholding happens here, it's still just modelling the data.
- *Inputs $\rightarrow$ Weighted Sum $\rightarrow$ Log-Odds $\rightarrow$ Sigmoid $\rightarrow$ Probability*

Why It's a Classification Algorithm (Practical View)

The model outputs a **probability** for each observation (e.g., 0.82 = 82% chance of class 1).

We apply a **decision threshold**, typically **0.5**, to assign a class:

- If $P > 0.5 \rightarrow \text{Class 1}$
- If $P \leq 0.5 \rightarrow \text{Class 0}$

This makes it act like a **binary classification algorithm**.

Different thresholds change **sensitivity vs specificity** — measured via **ROC/AUC** (coming next).

The goal becomes **accurate classification**, not just probability estimation.

Logistic Regression in Kaggle

- Good starting point in Kaggle competitions, before moving to complex models.
- Works well on clean, linearly separable data.
- Produces interpretable coefficients (**feature importance** insight)
- Efficient to train; handles large datasets quickly
- Serves as a benchmark to compare boosting models (XGBoost, LGBM, CatBoost)
- Current competition uses AUC as the metric — perfect for probability-based models like Logistic Regression.

Evaluating Logistic Regression – ROC & AUC

ROC (Receiver Operating Characteristic)

Shows how well the model distinguishes between Class 0 and 1.

AUC (Area Under Curve)

Single score that summarises the ROC curve.

Higher AUC = better model performance.

AUC = 1.0 → Perfect, AUC = 0.5 → Random guess.

TLDR

- **Linear vs Logistic:** *Same foundation, different interpretation*
- Linear Regression → predicts **continuous values**
- Logistic Regression → predicts **probabilities (0–1)**
- Add a **sigmoid** function → turn linear outputs into probabilities
- Acts as **regression in theory**, but **classification in practice**
- Evaluated using **ROC & AUC** to measure how well it separates classes
- Remains a **simple, fast, and strong baseline** for Kaggle competitions